



# MALLA REDDY UNIVERSITY

(As per Telangana Private Universities Act No. 13 of 2020 &  
G.O.Ms.No. 14, Higher Education (UE) Department)

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**SCHOOL OF ENGINEERING**  
**DEPARTMENT OF AI & ML (IV Year I Semester)**  
**Major Project Phase-I (MR23-1CS0273)**

**Date: 07/07/2026**

|                              |                                                                                                   |                               |
|------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Name of the Guide</b>     | Asst. Prof. Ch. Manasa                                                                            |                               |
| <b>Project Title</b>         | <b>OpenMentor: A Multi-Agent AI Framework for Personalized Open-Source Contributor Onboarding</b> |                               |
| <b>Batch Number</b>          | 15                                                                                                |                               |
| <b>Batch Student Details</b> | <b>Roll No</b>                                                                                    | <b>Student Name</b>           |
|                              | 2311CS020641                                                                                      | Surisetty Maruthi Ram Prakash |
|                              | 2311CS020642                                                                                      | Swargam Mani Mardhav          |
|                              | 2311CS020652                                                                                      | Terala Srikar Prasad          |

## Literature Survey

| S.no | Title                                                                                         | Author                                                                                                     | Published Year | Summary                                                                                                                                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Onboarding to Open Source Projects with Good First Issues: A Preliminary Analysis             | Horiguchi, H., Omori, I., Ohira, M.                                                                        | 2021           | Analyzed the real-world state of the "good first issue" (GFI) label mechanism on GitHub to understand how it affects newcomer onboarding, laying the empirical groundwork for automated GFI-recommendation models.                                                                |
| 2    | Supporting the Task-driven Skill Identification in Open Source Project Issue Tracking Systems | Santos, F., et al. (Northern Arizona University)                                                           | 2022           | Proposed automatically labelling open issues with API-domain skill categories mined from source code, so that newcomers can be matched to tasks that fit their existing skills rather than a generic difficulty tag.                                                              |
| 3    | SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering                    | Yang, J., Jimenez, C.E., Wettig, A., Lieret, K., Yao, S., Narasimhan, K., Press, O. (Princeton University) | 2024           | Built a custom agent-computer interface (ACI) that lets an LLM agent create/edit files, navigate whole repositories, and run tests, achieving state-of-the-art pass rates on SWE-bench and demonstrating that interface design is critical to agentic repository-level reasoning. |
| 4    | RepoHyper: Search-Expand-Refine on Semantic Graphs for Repository-Level Code Completion       | Phan, H.N., Phan, H.N., Nguyen, T.N., Bui, N.D.Q.                                                          | 2024           | Introduced the Repo-level Semantic Graph (RSG), a graph structure that encapsulates whole-repository context (files, classes, dependencies), and a search-expand-refine procedure so an LLM can retrieve precisely the structural context it needs.                               |

| <b>S.no</b> | <b>Title</b>                                                 | <b>Author</b>                                           | <b>Published Year</b> | <b>Summary</b>                                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------|---------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5           | Prioritization of Software Bugs Using Entropy-Based Measures | Kumari, M., Singh, R., Singh, V.B. (IIT Dhanbad, India) | 2025                  | Proposed a summary-entropy-based measure, combined with severity and summary weight, to capture the uncertainty and irregularity in reported bug data and more reliably predict bug priority in open-source repositories. |

**Guide**

**AD Mentor**

**AD Coordinator**

**Dean**